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By Amit Katiyar (MCA-JNU)

1. Exact value of $\sec\frac{7\pi}{12}$ **NIMCET 2007**
(a)$\sqrt{3}-1$ (b)$\sqrt{3}+1$
(c)$-\sqrt{2}(\sqrt{3}+1)$ (d)$2\sqrt{2}(\sqrt{3}+1)$

2. If $\cos\alpha+\cos\beta=a$, and $\sin\alpha+\sin\beta=b$ and θ is the arithmetic means between α and β, then $\sin 2\theta+\cos 2\theta$ is equal to **NIMCET 2008**
(a)$\frac{(a+b)^2}{(a^2+b^2)}$ (b)$\frac{(a-b)^2}{(a^2+b^2)}$
(c)$\frac{a^2-b^2}{a^2+b^2}$ (d)None of these

3. If $(1+\tan 1°)(1+\tan 2°)\ldots(1+\tan 45°)=2^n$, then the value of n is **NIMCET 2008**
(a)21 (b)22 (c)23 (d)24

4. The value of $\sin 12°\sin 48°\sin 54°$ is **NIMCET 2008**
(a)$\sin 30°$ (b)$\sin^2 30°$
(c)$\sin^3 30°$ (d)$\cos^3 30°$

5. If $A=\cos^2\theta+\sin^4\theta$, then for all value of θ **NIMCET 2009**
(a)$1\le A\le 2$ (b)$\frac{13}{16}\le A\le 1$
(c)$\frac{3}{4}\le A\le\frac{13}{16}$ (d)$\frac{3}{4}\le A\le 1$

6. The equation $\sin^4 x+\cos^4 x+\sin 2x+\alpha=0$ is solvable for **NIMCET 2009**
(a)$-\frac{1}{2}\le\alpha\le\frac{1}{2}$ (b)$-3\le\alpha\le 1$
(c)$-\frac{3}{2}\le\alpha\le\frac{1}{2}$ (d)$-1\le\alpha\le 1$

7. The value of $\sqrt{3}\cot 20°-4\cos 20°$ is **NIMCET 2010**
(a)1 (b)-1 (c)0 (d)None

8. The value of $\sin 30°\cos 45°+\cos 30°\sin 45°$ is **NIMCET 2011**
(a)$\frac{1-\sqrt{3}}{2}$ (b)$\frac{1+\sqrt{3}}{2\sqrt{2}}$ (c)$\frac{2}{\sqrt{3}}$ (d)$\frac{\sqrt{3}}{2}$

9. If $\tan\theta=\frac{b}{a}$, then the value of $a\cos 2\theta+b\sin 2\theta$ is **NIMCET 2011**
(a)b (b)a (c)$\frac{a}{b}$ (d)$\frac{a}{a+b}$

10. The value of $\frac{1-\tan^2 15°}{1+\tan^2 15°}$ is **NIMCET 2011**
(a)1 (b)$\sqrt{3}$ (c)$\frac{\sqrt{3}}{2}$ (d)2

11. If $\sin x$, $\cos x$ and $\tan x$ are in GP, then $\cot^6 x-\cot^2 x$ will be equal to **NIMCET 2011**
(a)2 (b)-1 (c)1 (d)0

12. If $\sin^2 x=1-\sin x$, then $\cos^4 x+\cos^2 x$ is equal to **NIMCET 2012**
(a)0 (b)1 (c)1/3 (d)-1

13. If $\cos(\alpha+\beta)=\frac{4}{5}$ and $\sin(\alpha-\beta)=\frac{5}{13}, 0<\alpha,\beta<\frac{\pi}{4}$, then $\tan(2\alpha)$ is equal to **NIMCET 2012**
(a)56/33 (b)63/65 (c)16/63 (d)33/56

14. If $A-B=\frac{\pi}{4}$, then $(1+\tan A)(1-\tan B)$ is equal to **NIMCET 2012**
(a)2 (b)1 (c)0 (d)3

15. Which of the following is correct? **NIMCET 2012**
(a) $\sin 1°>\sin 1$ (b) $\sin 1°<\sin 1$
(c) $\sin 1°=\sin 1$ (d) $\sin 1°=\frac{\pi}{180}\sin 1$

16. If $\sin(\pi\cos\theta)=\cos(\pi\sin\theta)$, then $\sin 2\theta$ is equal to **NIMCET 2012**
(a)$\pm\frac{3}{4}$ (b)$\pm\frac{1}{3}$ (c)$\pm\frac{1}{4}$ (d)$\pm\frac{4}{3}$

17. If $\sin x+a\cos x=b$, then what is the expression for $|a\sin x-\cos x|$ in terms of a and b **NIMCET 2013**
(a)$\sqrt{a^2-b^2-1}$ (b)$\sqrt{a^2+b^2-1}$
(c)$\sqrt{a^2+b^2+1}$ (d)$\sqrt{a^2-b^2+1}$

18. The value of $\tan\theta+2\tan 2\theta+4\tan 4\theta+8\cot 8\theta$ is **NIMCET 2013**
(a) $\cot\theta$ (b) $\tan\theta$
(c) $\sin\theta$ (d) $\cos\theta$

19. If $\sin x+\sin^2 x=1$, then $\cos^2 x+\cos^4 x$ is equal to **NIMCET 2013**
(a) 0 (b) 1 (c)-1 (d) 2

20. If $\tan\alpha=\frac{m}{m+1}$ and $\tan\beta=\frac{1}{2m+1}$, then $\alpha+\beta$ is equal to **NIMCET 2013**
(a)$\pi/3$ (b)$\pi/4$ (c)$\pi/6$ (d)π

21. The value of $\tan 1°\tan 2°\tan 3°\ldots\tan 89°$ is **NIMCET 2014**
(a)0 (b)$\frac{1}{\sqrt{2}}$ (c)1 (d)2

22. If $\sin x+a\cos x=b$, then $|a\sin x-\cos x|$ is **NIMCET 2014**
(a)$\sqrt{a^2+b^2+1}$ (b)$\sqrt{a^2-b^2+1}$
(c)$\sqrt{a^2+b^2-1}$ (d) None of these

23. If $\tan A-\tan B=x$ and $\cot B-\cot A=y$, then $\cot(A-B)$ is equal to **NIMCET 2014**
(a)$\frac{1}{x}+\frac{1}{y}$ (b)$\frac{1}{x}-\frac{1}{y}$ (c)$-\frac{1}{x}+\frac{1}{y}$ (d)$-\frac{1}{x}-\frac{1}{y}$

24. The value of $\sin 20°\sin 40°\sin 80°$ is **NIMCET 2014**
(a)$\frac{1}{2}$ (b)$\frac{\sqrt{3}}{2}$ (c)$\frac{\sqrt{3}}{8}$ (d)$\frac{1}{8}$

25. The value of $\tan\left(\frac{7\pi}{8}\right)$ is **NIMCET 2015**
(a)$1-\sqrt{2}$ (b)$1+\sqrt{2}$ (c)$\sqrt{2}+\sqrt{3}$ (d)$\sqrt{2}-\sqrt{3}$

26. If $P=\sin^{20}\theta+\cos^{48}\theta$, then the inequality that holds for the value of θ is **NIMCET 2015**
(a)$P\ge 1$ (b)$0<P\le 1$
(c)$1<P<3$ (d)$0\le P\le 1$

27. If $0<x<\pi$ and $\cos x+\sin x=\frac{1}{2}$, then the value of $\tan x$ is **NIMCET 2015**
(a)$\frac{4-\sqrt{7}}{3}$ (b)$\frac{-4+\sqrt{7}}{3}$ (c)$\frac{1+\sqrt{7}}{4}$ (d)$\frac{1-\sqrt{7}}{4}$

28. If $\cos\theta=\frac{5}{13},\frac{3\pi}{2}<\theta<2\pi$, then $\tan 2\theta$ is **NIMCET 2016**
(a)$\frac{-120}{119}$ (b)$\frac{-120}{169}$ (c)$\frac{119}{116}$ (d)$\frac{120}{119}$

29. The value of $\cos 20° + \cos 100° + \cos 140°$ is

NIMCET 2016

(a)0 (b)$\frac{1}{\sqrt{2}}$ (c)$\frac{1}{2}$ (d)1

30. If $\tan x = \frac{-3}{4}$ and $\frac{3\pi}{2} < x < 2\pi$, then the value of $\sin 2x$ is

NIMCET 2017

(a)7/25 (b)−7/25 (c)24/25 (d)−24/25

31. If $\cos\theta = \frac{4}{5}$, $\cos\phi = \frac{12}{13}$, with θ and ϕ both in the fourth quadrant, the value of $\cos(\theta + \phi)$ is

NIMCET 2017

(a)$-\frac{16}{65}$ (b)$-\frac{33}{65}$ (c)$\frac{33}{65}$ (d)$\frac{16}{65}$

32. The value of $\sin 36°$ is

NIMCET 2017

(a)$\frac{\sqrt{10+2\sqrt{5}}}{4}$ (b)$\frac{\sqrt{10-2\sqrt{5}}}{4}$

(c)$\frac{(\sqrt{5}+1)}{4}$ (d)$\frac{(\sqrt{5}-1)}{4}$

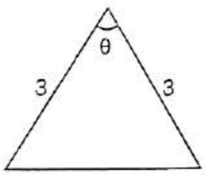
33. Express $(\cos 5x - \cos 7x)$ as a product of sines and cosines.

NIMCET 2017

(a)$2\cos 4x \cos x$ (b)$2\sin 4x \sin x$

(c)$2\sin 6x \sin x$ (d)$2\cos 6x \cos x$

34. What is the largest area of an isosceles triangle with two edges of length 3?



NIMCET 2017

(a)3 (b)3/2 (c)9 (d)9/2

35. The maximum value of $4\sin^2 x + 3\cos^2 x + \sin(x/2) + \cos(x/2)$ is

NIMCET 2017

(a)4 (b)$3+\sqrt{2}$ (c)9 (d)$4+\sqrt{2}$

36. In an acute angled ΔABC, the least value of $\sec A + \sec B + \sec C$ is

NIMCET 2018

(a)6 (b)8 (c)3 (d)2

37. Let $P = \{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta\}$ and $Q = \{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta\}$ be two sets. Then,

NIMCET 2018

(a)$P \subset Q \text{ and } Q - P$ (b)$P \not\subset Q$

(c)$Q \not\subset P$ (d)$P = Q$

38. If $\frac{\tan x}{2} = \frac{\tan y}{3} = \frac{\tan z}{5}$ and $x + y + z = \pi$, then the value of $\tan^2 x + \tan^2 y + \tan^2 z$ is

NIMCET 2018

(a)38/3 (b)38 (c)114 (d)None

39. If $\sin\theta = 3\sin(\theta + 2\alpha)$, then the value of $\tan(\theta + \alpha) + 2\tan\alpha$ is

NIMCET 2018

(a)3 (b)2 (c)−1 (d)0

40. If $A > 0, B > 0$ and $A + B = \frac{\pi}{6}$, then the minimum value of $\tan A + \tan B$ is

NIMCET 2019

(a)$\sqrt{3} - \sqrt{2}$ (b)$4 - 2\sqrt{3}$

(c)$\frac{2}{\sqrt{3}}$ (d)$2 - \sqrt{3}$

41. If $\frac{\tan x}{2} = \frac{\tan y}{3} = \frac{\tan z}{5}$ and $x + y + z = \pi$, then the value of $\tan^2 x + \tan^2 y + \tan^2 z$ is

NIMCET 2020

(a)38/3 (b)38 (c)114 (d)None

42. Find the value of $\sin 12° \sin 48° \sin 54°$. **NIMCET 2020**

(a)1/8 (b)1/6 (c)1/2 (d)1/4

43. If $\cos x = \tan y, \cot y = \tan z$ and $\cot z = \tan x$, then $\sin x$ is equal to **NIMCET 2020**

(a)$\frac{\sqrt{5}-1}{2}$ (b)$\frac{\sqrt{5}+1}{2}$ (c)$\frac{\sqrt{5}+1}{4}$ (d)$\frac{\sqrt{5}-1}{4}$

44. The value of $\tan\left(45 + \frac{\theta}{2}\right)$ is **NIMCET 2020**

(a)$\tan\theta - \sec\theta$ (b)$\tan\theta + \sec\theta$

(c)$\cot\theta - \sec\theta$ (d)$\cot\theta + \sec\theta$

45. The value of $\sin 10° \sin 50° \sin 70°$ is **NIMCET 2020**

(a)1/4 (b)1/2 (c)3/4 (d)1/8

46. The expression $\frac{\tan A}{1-\cot A} + \frac{\cot A}{1-\tan A}$ can be written as

NIMCET 2020

(a)$\sin A \cos A + 1$ (b)$\sec A \operatorname{cosec} A + 1$

(c)$\tan A + \cot A$ (d)$\sec A + \operatorname{cosec} A$

47. If $3\sin x + 4\cos x = 5$, then $6\tan\frac{x}{2} - 9\tan^2\frac{x}{2}$ is equal to

NIMCET 2020

(a)1 (b)3 (c)4 (d)6

48. If $|K| = 5$ and $0° < \theta < 360°$, then the number of different solutions of $3\cos\theta + 4\sin\theta = k$ is **NIMCET 2021**

(a)0 (b)1 (c)2 (d)infinite

49. If $a\cos\theta + b\sin\theta = 2$ and $a\sin\theta - b\cos\theta = 3$, then $a^2 + b^2$ is equal to **NIMCET 2021**

(a)13 (b)5 (c)10 (d)12

50. The value of $\tan 9° - \tan 27° - \tan 63° + \tan 81°$ is equal to **NIMCET 2021**

(a)5 (b)3 (c)4 (d)6

51. In a ΔABC, if $\tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} = k$, then k is always

NIMCET 2021

(a)> 1 (b)≥ 1 (c)$= 2$ (d)$= 1$

52. If $\operatorname{cosec}\theta - \cot\theta = 2$. Find $cosec\ \theta$. **NIMCET 2022**

(a)5/3 (b)3/5 (c)4/5 (d)5/4

53. If $\prod_{i=1}^{n} \tan(\alpha_i) = 1 \ \forall \ \alpha_i \ \epsilon \left[0, \frac{\pi}{2}\right]$ where $i = 1,2,3,\ldots,n$. Then maximum value of $\prod_{i=1}^{n} \sin\alpha_i$ **NIMCET 2023**

(a)$\frac{1}{2^n}$ (b)$\frac{1}{2^{n/2}}$ (c)1 (d)None

54. Largest value of $\cos^2\theta - 6\sin\theta\cos\theta + 3\sin^2\theta + 2$

NIMCET 2023

(a)4 (b)0

(c)$4 + \sqrt{10}$ (d)$4 - \sqrt{10}$

55. The maximum value of $\sin(x) + \sin(x+1)$ is $k\cos\frac{1}{2}$. Then the value of k is **NIMCET 2025**

(a) 1 (b) 2

(c) None of these (d) 3

56. If $\cos^2(10°)\cos(20°)\cos(40°)\cos(50°)\cos(70°) = \alpha + \frac{\sqrt{3}}{16}\cos(10°)$, then $3\alpha^{-1}$ is equal to **NIMCET 2025**

(a) $\frac{9}{32}$ (b) 64 (c) 16 (d) $\frac{9}{16}$

57. If $B = \sin^2 y + \cos^4 y$, then for all real y

NIMCET 2025

(a) $\frac{3}{4} \le B \le 1$ (b) $\frac{3}{4} \le B \le \frac{13}{16}$

(c) $\frac{13}{16} \le B \le 1$ (d) $1 \le B \le 2$

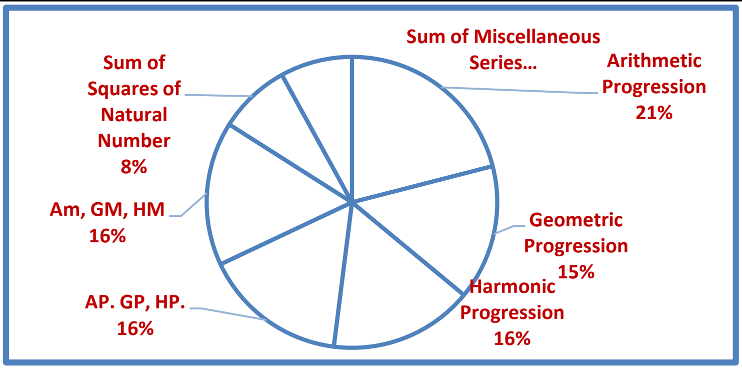
ANSWER KEY

1.	c	**2.**	d	**3.**	c	**4.**	c	**5.**	d
6.	c	**7.**	a	**8.**	b	**9.**	b	**10.**	c
11.	c	**12.**	b	**13.**	a	**14.**	a	**15.**	b
16.	a	**17.**	d	**18.**	a	**19.**	b	**20.**	b
21.	c	**22.**	b	**23.**	a	**24.**	c	**25.**	a
26.	b	**27.**	b	**28.**	d	**29.**	a	**30.**	d
31.	c	**32.**	b	**33.**	c	**34.**	d	**35.**	d
36.	a	**37.**	d	**38.**	a	**39.**	d	**40.**	b
41.	a	**42.**	a	**43.**	a	**44.**	b	**45.**	d
46.	b	**47.**	a	**48.**	c	**49.**	a	**50.**	c
51.	b	**52.**	d	**53.**	b	**54.**	c	**55.**	b
56.	a	**57.**	a						

Analysis 2007 to 2025
Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	01	NIMCET 2016	02
NIMCET 2008	03	NIMCET 2017	06
NIMCET 2009	02	NIMCET 2018	04
NIMCET 2010	01	NIMCET 2019	01
NIMCET 2011	04	NIMCET 2020	07
NIMCET 2012	05	NIMCET 2021	04
NIMCET 2013	04	NIMCET 2022	01
NIMCET 2014	04	NIMCET 2023	02
NIMCET 2015	03	NIMCET 2024	00
		NIMCET 2025	04

SUB TOPIC WISE ANALYSIS



Sub Topic wise Questions Asked in NIMCET

Sub Topic	Ques. Asked
Trigonometry Functions	27
Allied Angles	08
Transformation Formula	18
Multiple and Sub multiple Angles	19
Sum and Difference of Angles	02
Minimum Maximum Value	06
Compound Angles	20

SOLUTION

1. (c)

2. (d)

$\text{AM} = \frac{\alpha+\beta}{2} = \theta$

$\Rightarrow \frac{\cos\alpha+\cos\beta}{\sin\alpha+\sin\beta} = \frac{a}{b}$

$\Rightarrow \frac{2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)}{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)} = \frac{a}{b}$

$\Rightarrow \cot\left(\frac{\alpha+\beta}{2}\right) = \frac{a}{b} \Rightarrow \tan\theta = \frac{b}{a}$

$\Rightarrow \sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta} = \frac{2b/a}{1+b^2/a^2} = \frac{2ab}{a^2+b^2}$

and $\cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta} = \frac{1-b^2/a^2}{1+b^2/a^2} = \frac{a^2-b^2}{a^2+b^2}$

$\Rightarrow \sin 2\theta + \cos 2\theta = \frac{a^2-b^2+2ab}{a^2+b^2}$

3. (c)

$1 + \tan 1° = 1 + \tan(45° - 44°)$

$\tan(45° - 44°) = \frac{\tan 45° - \tan 44°}{1+\tan 45° \tan 44°} = \frac{1-\tan 44°}{1+\tan 44°}$

Hence, $(1+\tan 1°)(1+\tan 44°) = 2$

Similarly, $(1+\tan 2°)(1+\tan 43°) = 2$

$\Rightarrow (1+\tan 3°)(1+\tan 42°) = 2$

then $(1+\tan 1°)(1+\tan 2°)\ldots(1+\tan 43°)$

$(1+\tan 44°)(1+\tan 45°) = 2^n$

Hence, $2^{22} \times 2 = 2^n$

$2^n = 2^{23} \Rightarrow n = 23$

4. (c)

We know that $\sin(60° - \theta).\sin\theta.\sin(60° + \theta) = \frac{\sin 3\theta}{4}$

$= \frac{(\sin 48° \sin 12° \sin 72°)\sin 54°}{\sin 72°} = \frac{\sin 36° \sin 54°}{4\sin 72°}$

$= \frac{\sin 36° \sin(90° - 36°)}{4\sin 72°} = \frac{\sin 36° \cos 36°}{4\sin 72°}$

$= \frac{2\sin 36° \cos 36°}{8\sin 72°} = \frac{\sin 72°}{8\sin 72°} = \frac{1}{8} = \frac{1}{2^3} = \left(\frac{1}{2}\right)^3 = \sin^3 30°$

5. (d)

A$= \cos^2\theta + \sin^4\theta = \cos^2\theta + (\sin^2\theta)^2$

$= \cos^2\theta + (1-\cos^2\theta)^2$

$= \cos^2\theta + \cos^4\theta + 1 - 2\cos^2\theta$

$= \cos^4\theta - \cos^2\theta + 1$

$= \cos^4\theta - 2\left(\frac{1}{2}\right)\cos^2\theta + \frac{1}{4} - \frac{1}{4} + 1$

$= \left(\cos^2\theta - \frac{1}{2}\right)^2 + \frac{3}{4}$

$\Rightarrow 0 \le \cos^2\theta \le 1$

$\Rightarrow \frac{-1}{2} \le \cos^2\theta - \frac{1}{2} \le 1 - \frac{1}{2}$

$\Rightarrow 0 \le \left(\cos^2\theta - \frac{1}{2}\right)^2 \le \frac{1}{4}$

$\Rightarrow \frac{3}{4} \le \left(\cos^2\theta - \frac{1}{2}\right)^2 + \frac{3}{4} \le \frac{1}{4} + \frac{3}{4} \Rightarrow \frac{3}{4} \le A \le 1$

6. (c)

$\sin^4 x + \cos^4 x + \sin 2x + \alpha = 0$

$\Rightarrow 1 - 2\sin^2 x\cos^2 x + \sin 2x + \alpha = 0$

$[\because \sin^4 x + \cos^4 x = 1 - 2\sin^2 x\cos^2 x]$

$\Rightarrow 1 - \frac{\sin^2 2x}{2} + \sin 2x + \alpha = 0$

$\Rightarrow 2 - \sin^2 2x + 2\sin 2x + 2\alpha = 0$

Let sin 2x = t

$\Rightarrow t^2 - 2t - 2 - 2\alpha = 0$

$\Rightarrow (t-1)^2 - 3 - 2\alpha = 0$

$\Rightarrow (\sin 2x - 1)^2 - (3 + 2\alpha) = 0$

$\Rightarrow -1 \le \sin 2x \le 1 \Rightarrow -2 \le \sin 2x - 1 \le 0$

$\Rightarrow 0 \le (\sin 2x - 1)^2 \le 4 = 0 \le (3 + 2\alpha) \le 4$

$\Rightarrow -3 \le 2\alpha \le 1 \Rightarrow \frac{-3}{2} \le \alpha \le \frac{1}{2}$

7. (a)

$\sqrt{3}\cot 20^\circ - 4\cos 20^\circ$

$\Rightarrow \sqrt{3}\frac{\cos 20^\circ}{\sin 20^\circ} - 4\cos 20^\circ$

$\Rightarrow \frac{\sqrt{3}\cos 20^\circ - 4\cos 20^\circ \sin 20^\circ}{\sin 20^\circ}$

$\Rightarrow \frac{\sqrt{3}\cos 20^\circ - 2\sin 40^\circ}{\sin 20^\circ} \Rightarrow 2\left(\frac{\frac{\sqrt{3}}{2}\cos 20^\circ - \frac{2}{2}\sin 40^\circ}{\sin 20^\circ}\right)$

$\Rightarrow 2\frac{(\sin 60^\circ \cos 20^\circ - \sin 40^\circ)}{\sin 20^\circ}$

$\Rightarrow \frac{\sin(60^\circ + 20^\circ) + \sin(60^\circ - 20^\circ) - 2\sin 40^\circ}{\sin 20^\circ}$

$\Rightarrow \frac{\sin 80^\circ + \sin 40^\circ - 2\sin 40^\circ}{\sin 20^\circ} = \frac{\sin 80^\circ - \sin 40^\circ}{\sin 20^\circ}$

$\Rightarrow \frac{2\cos 60^\circ \sin 20^\circ}{\sin 20^\circ} = 2\cos 60^\circ = 2\left(\frac{1}{2}\right) = 1$

8. (b)

$\sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ$

$= \sin(30^\circ + 45^\circ) = \frac{1}{2} \times \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$

$= \frac{1+\sqrt{3}}{2\sqrt{2}} \Rightarrow \sin 75^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$

9. (b)

$a\cos 2\theta + b\sin 2\theta = a\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right) + b\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$

$\Rightarrow a\left(\frac{1-\frac{b^2}{a^2}}{1+\frac{b^2}{a^2}}\right) + b\left(\frac{2.\frac{b}{a}}{1+\frac{b^2}{a^2}}\right) = a\left(\frac{a^2-b^2}{a^2+b^2}\right) + b\left(\frac{2ab}{a^2+b^2}\right)$

$\Rightarrow \frac{a^3 - ab^2 + 2ab^2}{a^2+b^2} = a\frac{(a^2+b^2)}{a^2+b^2} = a$

10. (c)

$\frac{1-\tan^2 15^\circ}{1+\tan^2 15^\circ} = \frac{\cos^2 15^\circ - \sin^2 15^\circ}{\cos^2 15^\circ + \sin^2 15^\circ}$

$\Rightarrow \frac{\cos 2(15)}{1} = \cos 30^\circ = \frac{\sqrt{3}}{2}$

11. (c) $\cos^2 x = \sin x \cdot \tan x$

$\Rightarrow \cos^2 x = \frac{\sin^2 x}{\cos x} \Rightarrow \cos x = \frac{\sin^2 x}{\cos^2 x}$

$\Rightarrow \cos x = \tan^2 x \Rightarrow \sec x = \cot^2 x$

$\Rightarrow \cot^6 x - \cot^2 x = (\cot^2 x)^3 - \cot^2 x$

$\Rightarrow \sec^3 x - \sec x \Rightarrow \sec x(\sec^2 x - 1)$

$\Rightarrow \sec x(\tan^2 x) = \sec x\cos x = 1$

12. (b)

If $\sin^2 x = 1 - \sin x$

$\Rightarrow \sin x = 1 - \sin^2 x \Rightarrow \sin x = \cos^2 x$

Hence, $\cos^4 x + \cos^2 x = (\cos^2 x)^2 + \cos^2 x$

$= \sin^2 x + \cos^2 x = 1$

13. (a)

$\tan 2\alpha = \tan\{(\alpha+\beta) + (\alpha-\beta)\}$

$\tan 2\alpha = \frac{\tan(\alpha+\beta) + \tan(\alpha-\beta)}{1 - \tan(\alpha+\beta) \times \tan(\alpha-\beta)}$

$\cos(\alpha+\beta) = \frac{4}{5}, \sin(\alpha-\beta) = \frac{5}{13}$

Hence, $\tan(\alpha+\beta) = \frac{3}{4}, \tan(\alpha-\beta) = \frac{5}{12}$

$\tan 2\alpha = \frac{\frac{3}{4}+\frac{5}{12}}{1-\frac{15}{48}} = \frac{56}{33}$

$\Rightarrow \tan 2\alpha = \frac{56}{33}$

14. (a)

$A - B = \frac{\pi}{4}$

$\tan(A-B) = \tan\frac{\pi}{4} \Rightarrow \frac{\tan A - \tan B}{1 + \tan A\tan B} = 1$

$\tan A - \tan B - \tan A\tan B = 1$

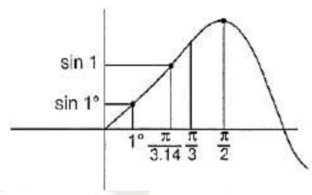
Now, $(1 + \tan A)(1 - \tan B)$

$= 1 - \tan B + \tan A - \tan A\tan B$

$= 1 + (\tan A - \tan B - \tan A\tan B) = 1 + 1 = 2$

15. (b)

We should know, $1 \cong \frac{\pi}{3.14}$



From, graph we can say sin 1 >sin 1^0

16. (a)

$\sin(\pi\cos\theta) = \sin\left(\frac{\pi}{2} - \pi\sin\theta\right) \Rightarrow \pi\cos\theta = \frac{\pi}{2} - \pi\sin\theta$

$\Rightarrow \cos\theta = \frac{1}{2} - \sin\theta \Rightarrow \sin\theta + \cos\theta = \frac{1}{2}$

$\cos\left(\frac{\pi}{2} - \pi\cos\theta\right) = \cos(\pi\sin\theta)$

$\frac{\pi}{2} - \pi\cos\theta = \pm\pi\sin\theta$

$\frac{\pi}{2} - \pi\cos\theta = -\pi\sin\theta \Rightarrow \frac{1}{2} = \cos\theta - \sin\theta$

$1 - \sin 2\theta = \frac{1}{4} \Rightarrow \sin 2\theta = \frac{3}{4}$ and $\frac{\pi}{2} - \pi\cos\theta = \pi\sin\theta$

$\Rightarrow (\sin\theta + \cos\theta) = \frac{1}{2} \Rightarrow \sin\theta + \cos\theta = \frac{1}{2}$

$1 + \sin 2\theta = \frac{1}{4} \Rightarrow \sin 2\theta = -\frac{3}{4}$

17. (d)

$\sin x + a\cos x = b \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Let $|a\sin x - \cos x| = k$ (say) ....(ii)

Squaring and adding Eqs. (i) and (ii)

$\Rightarrow \sin^2 x + a^2\cos^2 x + 2a\sin x\cos x$

$+\cos^2 x + a^2\sin^2 x - 2a\sin x\cos x = b^2 + k^2$

$\Rightarrow 1 + a^2 = b^2 + k^2 \Rightarrow k^2 = 1 + a^2 - b^2$

$\Rightarrow\ k = \pm\sqrt{a^2 - b^2 + 1}$

$\because\ |a\sin x - \cos x| = k = \sqrt{a^2 - b^2 + 1}$

18. (a)

$\tan A - \cot A = \frac{\sin^2 A - \cos^2 A}{\cos A \sin A} = \frac{-2\cos 2A}{2\cos A \sin A}$

$= \frac{-2\cos 2A}{\sin 2A} = -2\cot 2A$

Add and subtract $\cot\theta$ in given equation.

$= \ \tan\theta - \cot\theta + 2\tan 2\theta + 4\tan 4\theta + 8\cot 8\theta + \cot\theta$

$= -2\cot 2\theta + 2\tan 2\theta + 4\tan 4\theta + 8\cot 8\theta + \cot\theta$

$= 2(\tan 2\theta - \cot 2\theta) + 4\tan 4\theta + 8\cot 8\theta + \cot\theta$

$= 2(-2\cot 4\theta) + 4\tan 4\theta + 8\cot 8\theta + \cot\theta$

$= 4(\tan 4\theta - \cot 4\theta) + 8\cot 8\theta + \cot\theta$

$= 4(-2\cot 8\theta) + 8\cot 8\theta + \cot\theta = \cot\theta$

Short Trick:

Let $\theta = 30°$

$\Rightarrow \tan 30° + 2\tan 60° + 4\tan 120° + 8\cot 240°$

$\Rightarrow \tan 30° + 2\tan 60° - 4\tan 60° + 8\tan 30°$

$\Rightarrow \frac{1}{\sqrt{3}} + 2\sqrt{3} - 4\sqrt{3} + \frac{8}{\sqrt{3}}$

$\Rightarrow \frac{9}{\sqrt{3}} - 2\sqrt{3} = \frac{9-2(3)}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3} = \cot 30°$

19. (b)

$\sin x + \sin^2 x = 1 \Rightarrow \sin x = \cos^2 x$

$\cos^2 x + \cos^4 x = (\sin x) + (\sin x)^2 = 1$

20. (b)

$\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha \tan\beta}$

$\Rightarrow \tan(\alpha+\beta) = \frac{\frac{m}{m+1} + \frac{1}{2m+1}}{1 - \frac{m}{m+1}\left(\frac{1}{2m+1}\right)} = \frac{m(2m+1)+m+1}{(m+1)(2m+1)-m}$

$= \frac{2m^2+m+m+1}{2m^2+3m+1-m} = 1$

$\Rightarrow\ \alpha + \beta = \tan^{-1}(1) = \frac{\pi}{4}$

21. (c)

$\tan 1° \tan 2° \tan 3° \ldots \tan 45° \ldots \tan 87° \tan 88° \tan 89°$

$= \tan 1° \tan 2° \tan 3° \ldots (1) \ldots \cot 3° \cot 2° \cot 1 = 1$

22. (b) Refer to solution 16.

23. (a) $\cot(A - B) = \frac{1}{\tan(A-B)}$

$\because \tan A - \tan B = x$ and $\frac{1}{\tan B} - \frac{1}{\tan A} = y$

$\Rightarrow \frac{\tan A - \tan B}{\tan A \tan B} = y \Rightarrow \tan A \cdot \tan B = \frac{x}{y}$

So, $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} = \frac{x}{1 + x/y} = \frac{xy}{x+y}$

$\Rightarrow \cot(A - B) = \frac{x+y}{xy} = \frac{1}{x} + \frac{1}{y}$

24. (c)

$\sin 20° \sin 40° \sin 80°$

$= \sin(60° - 20°)\sin 20° \sin(60° + 20°)$

$= \frac{\sin 3(20°)}{4} = \frac{\sin 60°}{4} = \frac{\sqrt{3}}{2\times 4} = \frac{\sqrt{3}}{8}$

25. (a)

$\tan\left(\frac{7\pi}{8}\right) = \tan\left(\pi - \frac{\pi}{8}\right) = -\tan\frac{\pi}{8}$

We know that

$\tan\frac{\pi}{8} = \sqrt{2} - 1 \Rightarrow -\tan\frac{\pi}{8} = -(\sqrt{2} - 1) = 1 - \sqrt{2}$

26. (b)

$P = \sin^{20}\theta + \cos^{48}\theta$

$P = \sin^{20}\theta + \cos^{48}\theta > 0, \forall\theta$

Now, $P = \sin^{20}\theta + (1 - \sin^2\theta)^{24}$

$0 \le \sin^2\theta \le 1$

$0 \le \sin^{20}\theta \le 1$

$\Rightarrow\ 0 \le \sin^{18}\theta \le 1$

Now, $P = \sin^{20}\theta + (1 - \sin^2\theta)^{24}$

Hence, $0 \le \sin^{20}\theta \le \sin^2\theta$

Similarly, $0 \le \cos^{46}\theta \cdot \cos^2\theta \le \cos^2\theta$

Add Eqs. (i) and (ii),

$\Rightarrow 0 < \sin^{20}\theta + \cos^{48}\theta \le \sin^2\theta + \cos^2\theta$

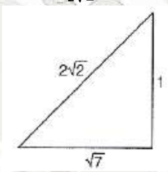
$\Rightarrow 0 < \sin^{20}\theta + \cos^{48}\theta \le 1$

$\Rightarrow 0 < \mathrm{P} \le 1$

27. (b)

$\cos x + \sin x = \frac{1}{2}[0 < x < \pi]$

$\Rightarrow \sqrt{2}\left[\frac{1}{\sqrt{2}}\cos x + \frac{1}{\sqrt{2}} sin x\right] = \frac{1}{2} \Rightarrow sin\left(x + \frac{\pi}{4}\right) = \frac{1}{2\sqrt{2}}$



$\Rightarrow \tan\left(x + \frac{\pi}{4}\right) = \frac{1}{\sqrt{7}} \Rightarrow \frac{\tan x + \tan\frac{\pi}{4}}{1 - \tan x \tan\frac{\pi}{4}} = \frac{\tan x + 1}{1 - \tan x} = \frac{1}{\sqrt{7}}$

$\Rightarrow\ \sqrt{7}\tan x + \sqrt{7} = 1 - \tan x$

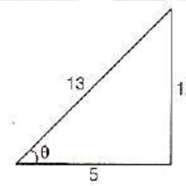
$\Rightarrow\ (1 + \sqrt{7})\tan x = 1 - \sqrt{7}$

$\Rightarrow\ \tan x = \frac{1-\sqrt{7}}{1+\sqrt{7}} = \frac{(1-\sqrt{7})^2}{1-7}$

$= \frac{1+7-2\sqrt{7}}{-6} = \frac{2(4-\sqrt{7})}{-6} = \frac{\sqrt{7}-4}{3}$

28. (d)

$\cos\theta = \frac{5}{13}; \frac{3\pi}{2} < \theta < 2\pi \Rightarrow 3\pi < 2\theta < 4\pi$



$\Rightarrow \tan\theta = \frac{12}{5}$

$\because\ \theta \in 4^{\text{th}}$ quad $\ \Rightarrow \tan\theta = \frac{-12}{5}$

$\Rightarrow \tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$

$= \frac{2\left(\frac{-12}{5}\right)}{1 - \left(\frac{-12}{5}\right)^2} = \frac{-24(5)}{25-144}$

$= \frac{-120}{-119} = \frac{120}{119}$

29. (a) $\cos 20° + \cos 100° + \cos 140°$

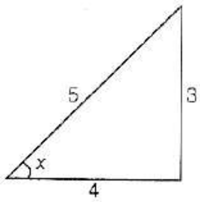
$\Rightarrow\ 2\cos\left(\frac{100+20}{2}\right)\cos\left(\frac{100-20}{2}\right) + \cos 140°$

$\Rightarrow\ 2\left(\frac{1}{2}\right)\cos 40° + \cos(180° - 40°)$

$\Rightarrow\ \cos 40° - \cos 40° = 0$

30. (d)

$\tan x = \frac{-3}{4}, x \in IV$ (quad)

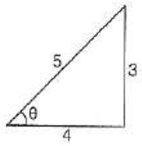


$\Rightarrow \sin x = -\frac{3}{5} \Rightarrow \cos x = \frac{4}{5}$

$\Rightarrow \sin 2x = 2\sin x\cos x = -2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{-24}{25}$

31. (c)

$\theta, \phi \in$ IVth quad $\cos\theta = \frac{4}{5}$



$\Rightarrow \ \sin\theta = -\frac{3}{5} \Rightarrow \cos\phi = \frac{12}{13}$

$\Rightarrow \ \sin\phi = -\frac{5}{13}$

$\Rightarrow \cos(\theta+\phi) = \cos\theta\cos\phi - \sin\theta\sin\phi$

$= \left(\frac{4}{5}\right)\left(\frac{12}{13}\right) - \left(-\frac{3}{5}\right)\left(-\frac{5}{13}\right) = \frac{48-15}{65} = \frac{33}{65}$

32. (b)

$\sin 36^\circ = \sqrt{1-\cos^2 36^\circ} = \sqrt{1-\left(\frac{\sqrt{5}+1}{4}\right)^2}$

$= \sqrt{\frac{16-6-2\sqrt{5}}{16}} = \frac{\sqrt{10-2\sqrt{5}}}{4}$

33. (c)

$2\sin\left(\frac{5x+7x}{2}\right)\sin\left(\frac{7x-5x}{2}\right) \Rightarrow 2\sin 6x\sin x$

34. (d)

Using area of triangle= $\frac{1}{2}$bc $\sin\theta$

$\Delta = \frac{1}{2}(3)(3)\sin\theta = \frac{9}{2}\sin\theta$

$\because \ -1 \le \sin\theta \le 1$

$\Rightarrow \ \frac{9}{2}\sin 90^\circ = \frac{9}{2} =$ maximum area

35. (d)

$4\sin^2 x + 3\cos^2 x + \sin\left(\frac{x}{2}\right) + \cos\left(\frac{x}{2}\right)$

$\Rightarrow \ 3 + \sin^2 x + \sin\frac{x}{2} + \cos\frac{x}{2}$

$\Rightarrow$ max. value of $\sin\frac{x}{2} + \cos\frac{x}{2} = \sqrt{2}$

$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2} \Rightarrow x = 90^\circ$

$\Rightarrow \ 3 + \sin^2 90^\circ + \sin 45^\circ + \cos 45^\circ = 4 + \sqrt{2}$

36. (a)

$f(x) = \sec A + \sec B + \sec C$

$\because$ For any triangle $\Rightarrow \cos A + \cos B + \cos C = \frac{3}{2}$

$A + B + C = \pi$ $\qquad \because A, B, C \in \mathrm{I}^{\text{st}}$ quad

so all angles will be positive $\Rightarrow$ AM $\ge$ GM

$\Rightarrow \ \frac{\cos A + \cos B + \cos C}{3} \ge (\cos A\cos B\cos C)^{1/3}$

$\Rightarrow \ \left(\frac{3/2}{3}\right)^3 \ge \cos A\cos B\cos C$

$\Rightarrow \ \cos A\cos B\cos C \le 1/8 \Rightarrow \sec A\sec B\sec C \le 8$

$\frac{\sec A + \sec B + \sec C}{3} \ge (\sec A\sec B\sec C)^{1/3}$

$\Rightarrow \ \sec A + \sec B + \sec C \ge 3 \times 2$

$\Rightarrow \ \sec A + \sec B + \sec C \ge 6$

37. (d)

$P\!: \ \sin\theta - \cos\theta = \sqrt{2}\cos\theta$

$\Rightarrow \sin\theta = (\sqrt{2}+1)\cos\theta \Rightarrow \tan\theta = \sqrt{2}+1$

$Q\!: \sin\theta + \cos\theta = \sqrt{2}\sin\theta \Rightarrow \cos\theta = (\sqrt{2}-1)\sin\theta$

$\Rightarrow \tan\theta = \frac{1}{\sqrt{2}-1} = \sqrt{2}+1 \Rightarrow P = Q$

38. (a)

$\frac{\tan x}{2} = \frac{\tan y}{3} = \frac{\tan z}{5} = k$

$\Rightarrow \ \tan x = 2k, \tan y = 3k$ and $\tan z = 5k$

$x + y = \pi - z$

$\Rightarrow \ \tan(x+y) = -\tan z \Rightarrow \frac{\tan x + \tan y}{1 - \tan x\tan y} = -\tan z$

$\Rightarrow \ \tan x + \tan y + \tan z = \tan x\tan y\tan z$

$\Rightarrow 2k + 3k + 5k = (2k)(3k)(5k)$

$\Rightarrow 10k = 30k^3 \Rightarrow k^2 = \frac{1}{3}$

and $(\tan x)^2 + (\tan y)^2 + (\tan z)^2 = 38k^2 = \frac{38}{3}$

39. (d)

$3\sin(\theta + \alpha + \alpha) = \sin(\theta + \alpha - \alpha)$

$\Rightarrow 3[\sin(\theta+\alpha)\cos\alpha + \cos(\theta+\alpha)\sin\alpha]$

$= \sin(\theta+\alpha)\cos\alpha - \cos(\theta+\alpha)\sin\alpha$

$\Rightarrow \ \sin(\theta+\alpha)\cos\alpha = -2\cos(\theta+\alpha)\sin\alpha$

$\Rightarrow 2\tan\alpha + \tan(\theta+\alpha) = 0$

Alternate solution

$\sin\theta = 3\sin(\theta + 2\alpha)$

Let $\theta = \alpha = 0$

$\Rightarrow 0 = 0$; verfied. $\Rightarrow \tan(\alpha+\theta) + 2\tan\alpha = 0$

40. (b)

$\tan(A+B) = \tan\frac{\pi}{6} \Rightarrow \frac{\tan A + \tan B}{1 - \tan A\tan B} = \frac{1}{\sqrt{3}}$

$\Rightarrow 1 - \tan A\tan B = \sqrt{3}(\tan A + \tan B)$

$\Rightarrow 1 - \sqrt{3}(\tan A + \tan B) = \tan A\tan B$

$\because A, B > 0 \Rightarrow$ AM $\ge$ GM

$\Rightarrow \ \frac{\tan A + \tan B}{2} \ge (\tan A\tan B)^{1/2}$

Let $\tan A + \tan B = x$

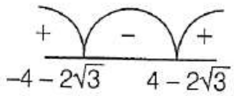
$\Rightarrow \ \frac{x}{2} \ge (1 - \sqrt{3}x)^{1/2}$

$\Rightarrow \ x^2 \ge 4 - 4\sqrt{3}x$

$\Rightarrow \ x^2 + 4\sqrt{3}x - 4 \ge 0$

$\Rightarrow x = \frac{-4\sqrt{3} \pm \sqrt{48+16}}{2} = \frac{-4\sqrt{3} \pm 8}{2}$

$\Rightarrow x = -2\sqrt{3} \pm 4$



$\Rightarrow \tan A + \tan B > 0 \Rightarrow \tan A + \tan B \in [4, -2\sqrt{3}, \infty]$

$\Rightarrow$ Min value of $\tan A + \tan B = 4 - 2\sqrt{3}$

41. (a) Refer to solution 37.

42. (a)

We should know $\sin(60^\circ - \theta).\sin\theta.\sin(60^\circ + \theta) = \frac{\sin 3\theta}{4}$

$\Rightarrow \frac{(\sin 48^\circ \sin 12^\circ \sin 72^\circ)\sin 54^\circ}{\sin 72^\circ}$

$= \frac{\sin 36^\circ \sin 54^\circ}{4\sin 72^\circ} = \frac{\sin 36^\circ \sin(90^\circ - 36^\circ)}{4\sin 72^\circ}$

$= \frac{\sin 36^\circ \cos 36^\circ}{4\sin 72^\circ} = \frac{2\sin 36^\circ \cos 36^\circ}{8\sin 72^\circ} = \frac{\sin 72^\circ}{8\sin 72^\circ} = \frac{1}{8}$

43. (a)

$\cos x = \tan y = \frac{1}{\tan z} = \tan x \Rightarrow \cos x = \tan x = \frac{\sin x}{\cos x}$

$\Rightarrow \sin x = \cos^2 x \Rightarrow \sin x = 1 - \sin^2 x$

$\Rightarrow \sin^2 x + \sin x - 1 = 0 \Rightarrow \sin x = \frac{-1\pm\sqrt{1+4}}{2} = \frac{-1\pm\sqrt{5}}{2}$

From option $\Rightarrow \frac{\sqrt{5}-1}{2}$

44. (b)

$\tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right) = \frac{\cos\frac{\theta}{2}+\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}-\sin\frac{\theta}{2}}$

$\Rightarrow \frac{\left(\cos\frac{\theta}{2}+\sin\frac{\theta}{2}\right)^2}{\cos^2\frac{\theta}{2}-\sin^2\frac{\theta}{2}}$

$\Rightarrow \frac{1+\sin\theta}{\cos\theta} = \sec\theta + \tan\theta$

45. (d)

$\sin(60^\circ - 10^\circ) . \sin 10^\circ . \sin(60^\circ + 10^\circ)$

$\Rightarrow \frac{\sin 3(10)}{4} = \frac{1}{8}$

46. (b)

$\frac{\tan A}{1-\cot A} + \frac{\cot A}{1-\tan A}$

$\Rightarrow \frac{\sin A/\cos A}{1-\cos A/\sin A} + \frac{\cos A/\sin A}{1-\sin A/\cos A}$

$\Rightarrow \frac{\sin^2 A}{\cos A(\sin A-\cos A)} + \frac{\cos^2 A}{\sin A(\cos A-\sin A)}$

$\Rightarrow \frac{\sin^3 A-\cos^3 A}{\sin A\cos A(\sin A-\cos A)}$

$\Rightarrow \frac{(\sin A-\cos A)(1+\sin A\cos A)}{(\sin A-\cos A)(\sin A\cos A)} \Rightarrow \operatorname{cosec} A\sec A + 1$

47. (a)

$3\sin x + 4\cos x = 5$

$3\left(\frac{2\tan x/2}{1+\tan^2 x/2}\right) + 4\left(\frac{1-\tan^2 x/2}{1+\tan^2 x/2}\right) = 5$

Let $\tan\frac{x}{2} = t$

Solving Eq. (i), we get $9t^2 - 6t + 1 = 0$

Hence, $(3t-1)^2 = 0$

$\Rightarrow t = \frac{1}{3}$

The value of $6\tan\frac{x}{2} - 9\tan^2\frac{x}{2} = 6 \times \frac{1}{3} - 9 \times \frac{1}{9} = 1$

48. (c)

$3\cos x + 4\sin x = 5$ and $|K| = 5$

$\Rightarrow 3\cos x + 4\sin x = \pm 5$

$\Rightarrow K = \pm 5$

$\because -\sqrt{3^2+4^2} \le 4\sin x + 3\cos x \le \sqrt{3^2+4^2}$

$\Rightarrow -5 \le 4\sin x + 3\cos x \le 5$

$\Rightarrow$ 2 different solutions

49. (a)

$a\cos\theta + b\sin\theta = 2$ .... (i)

$a\sin\theta - b\cos\theta = 3$ ....(ii)

Squaring and adding Eqs. (i) and (ii)

$\Rightarrow a^2 + b^2 = 4 + 9 = 13$

50. (c)

$\tan 9^\circ + \tan 81^\circ - (\tan 27^\circ + \tan 63^\circ)$

$\Rightarrow \tan 9^\circ + \cot 9^\circ - (\tan 27^\circ + \cot 27^\circ)$

$\Rightarrow \frac{\sin 9^\circ}{\cos 9^\circ} + \frac{\cos 9^\circ}{\sin 9^\circ} - \left(\frac{\sin 27^\circ}{\cos 27^\circ} + \frac{\cos 27^\circ}{\sin 27^\circ}\right)$

$\Rightarrow \frac{1}{\sin 9^\circ \cos 9^\circ} - \left(\frac{1}{\sin 27^\circ \cos 27^\circ}\right)$

$\Rightarrow \frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ} = \frac{2}{\sin 18^\circ} - \frac{2}{\cos 36^\circ}$

$\Rightarrow 2\left[\frac{4}{\sqrt{5}-1} - \frac{4}{\sqrt{5}+1}\right] = 8\left[\frac{\sqrt{5}+1-\sqrt{5}+1}{5-1}\right] \Rightarrow \frac{16}{4} = 4$

51. (b)

$\frac{A+B}{2} = \frac{\pi}{2} - \frac{C}{2} \Rightarrow \tan\left(\frac{A+B}{2}\right) = \tan\left(\frac{\pi}{2} - \frac{C}{2}\right)$

$\Rightarrow \tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = 1$

Let $\tan\frac{A}{2} = x, \tan\frac{B}{2} = y, \tan\frac{C}{2} = z$

Hence, $xy + yz + zx = 1$

We should know

$(x-y)^2 \ge 0 \Rightarrow x^2 + y^2 \ge 2xy$

$(y-z)^2 \ge 0 \Rightarrow y^2 + z^2 \ge 2yz$

$(z-x)^2 \ge 0 \Rightarrow z^2 + x^2 \ge 2zx$

Add these three equations, we get

$x^2 + y^2 + z^2 \ge xy + yz + zx$

Hence, $x^2 + y^2 + z^2 \ge 1$

$\Rightarrow \tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} \ge 1$

Alternate Solution

For $A = B = C = \frac{\pi}{3}$

$= \tan^2 30^\circ + \tan^2 30^\circ + \tan^2 30^\circ = 3\left(\frac{1}{\sqrt{3}}\right)^2 = 1$

For $A = 30^\circ, B = 60^\circ, C = 90^\circ$

$= \tan^2 15^\circ + \tan^2 30^\circ + \tan^2 45^\circ$

$= (2-\sqrt{3})^2 + \frac{1}{3} + 1 = 4 + 3 - 4\sqrt{3} + \frac{4}{3} = 7 + \frac{4}{3} - 4\sqrt{3}$

$= \frac{25}{3} - 4\sqrt{3} = \frac{25}{3} - 4\sqrt{3} > 0$

So, $\tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} \ge 1$

52. (d) $\operatorname{cosec}\theta - \cot\theta = 2$

$\operatorname{cosec}\theta + \cot\theta = \frac{1}{2}$

Adding Eqs. (i) and (ii) we get $2\operatorname{cosec}\theta = \frac{5}{2}$

Hence, $\operatorname{cosec}\theta = \frac{5}{4}$

53. (b)

$\sin\alpha_1 \sin\alpha_2 \sin\alpha_3 \dots \sin\alpha_n = \cos\alpha_1 \cos\alpha_2 \dots \cos\alpha_n$

Multiplying both sides by $\sin\alpha_1 \sin\alpha_2 \sin\alpha_3 \dots \sin\alpha_n$

$\Rightarrow \sin^2\alpha_1 \sin^2\alpha_2 \sin^2\alpha_3 \dots \sin^2\alpha_n$

$= (\sin\alpha_1\cos\alpha_1)(\sin\alpha_2\cos\alpha_2)\dots(\sin\alpha_n\cos\alpha_n)$

$\Rightarrow \sin^2\alpha_1 \sin^2\alpha_2 \sin^2\alpha_3 \dots \sin^2\alpha_n$

$= \frac{1}{2^n}(\sin 2\alpha_1)(\sin 2\alpha_2)\dots(\sin 2\alpha_n)$

As we know maximum value of $\sin\theta$ is 1

$\Rightarrow \sin^2\alpha_1 \sin^2\alpha_2 \sin^2\alpha_3 \dots \sin^2\alpha_n \le \frac{1}{2^n}$

$\Rightarrow \sin\alpha_1 \sin\alpha_2 \sin\alpha_3 \dots \sin\alpha_n \le \frac{1}{2^{n/2}}$

Hence, maximum value of $\prod_{i=1}^{n} \sin\alpha_i = \frac{1}{2^{n/2}}$

54. (c)

$\cos^2\theta - 6\sin\theta\cos\theta + 3\sin^2\theta + 2$

$= 2\sin^2\theta - 6\sin\theta\cos\theta + 3$

$= 3 - 3\sin 2\theta + [1 - \cos 2\theta]$

$= 4 - [3\sin 2\theta + \cos 2\theta]$

$-\sqrt{10} \le 3\sin 2\theta + \cos 2\theta \le \sqrt{10}$

For maximum value of given expression $3\sin 2\theta + \cos 2\theta$
It should be minimum.
Hence, maximum value if $4 + \sqrt{10}$.

55. (b)

1. Use the sum-to-product identity:

$$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right) cos\left(\frac{A-B}{2}\right)$$

Apply with A = x, B = $x + 1$:

$$\sin x + \sin(x+1) = 2\sin\left(x + \frac{1}{2}\right) cos\left(\frac{1}{2}\right)$$

2. The factor $\cos\frac{1}{2}$ is a constant (assume angles in radians). The other factor sin $\left(x + \frac{1}{2}\right)$ varies between -1 and 1. Its maximum value is 1.
3. The maximum of the whole expression is therefore

$$2.1.\cos\frac{1}{2} = 2cos\frac{1}{2}$$

This maximum is attainable: choose x so that $x + \frac{1}{2} = \frac{\pi}{2}, i.e., \frac{\pi}{2} - \frac{1}{2}$; then sin $\left(x + \frac{1}{2}\right) = 1$.

4. Given the maximum is k cos $\frac{1}{2}$, we have

$$k\cos\frac{1}{2} = 2\cos\frac{1}{2} \Rightarrow k = 2.$$

56. (a)

ratio

$cos^2(10°)cos(20°)cos(40°)cos(50°)cos(70°)$

= $cos^2(10°)cos(20°)cos(40°)sin(50°)sin(70°)$

= $\frac{1}{4}cos^3(10°)sin(40°)$

= $cos^4(10°)sin(10°)cos(20°)$

This reduces to the form

$$\alpha + \frac{\sqrt{3}}{16}cos(10°),$$

Hence,

$$3\alpha - 1 = \frac{9}{32}$$

57. (a)

To find the range of B = $\sin^2 y + \cos^4 y$ we can express B entirely in terms of $\sin^2 y$.

1. Substitute $\cos^2 y = 1 - \sin^2 y$ into the expression for B
   B = $\sin^2 y + (1 - \sin^2 y)^2$
2. Let x = $\sin^2 y$.
   Since y is a real number, 0< $\sin^2 y \leq 1$, so $0 \leq x < 1$
   B = $x + (1-x)^2$
   B = $x + 1 - 2x + x^2$
   B = $x^2 - x + 1$
3. To find the range of this quadratic function for $0 \leq x \leq 1$, we can find the vertex and evaluate the function at the endpoints.
   The x-coordinate of the vertex is given by x = - (-1)/(2.1).
   Since 1/2 is within the interval, the minimum value of B occurs at x = 1/2.
   Minimurn value of
   B = $(1/2)^2$ - (1/2) + 1 = 1/4 - 1/2 + 1 = 1/4 - 2/4 + 4/4 - 3/4
4. Now, evaluate B at the endpoints of the interval for x:
   At x = 0 (i.e, $\sin^2 y = 0$) : B = 0^2 - 0 + 1 = 1.
   At x = 1 (i.e., $\sin^2 y = 1$) : B = 1^2 - 1 + 1 = 1.
5. Comparing the values, the maximum value of B is 1
   Therefore, the range of B is $3/4 \leq B \leq 1$.
   The final answer is (a).